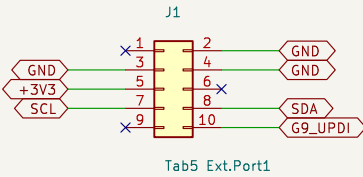
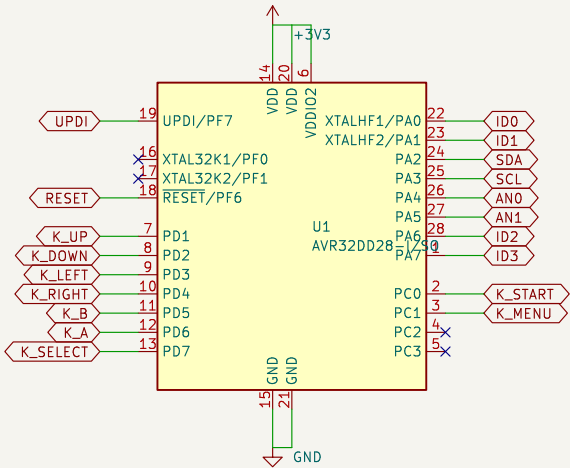


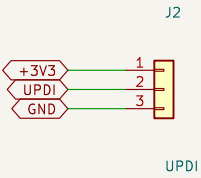
MCU – AVR32DD28 (SOIC–28) running firmware_avrdd (I2C client 0x6D, RetroPad Ext.Port1) – verify pin 1 orientation before fab

Buttons – one MCU pin each, switch to GND (internal pull–up); slot table in firmware_avrdd/pinmap.h

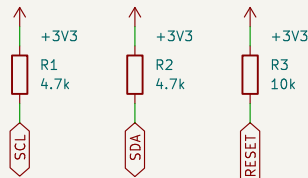
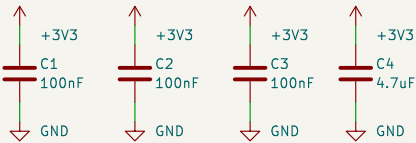
PD1–PD7, PC0–PC3, PF0–PF1: one button per pin (internal pull–ups)



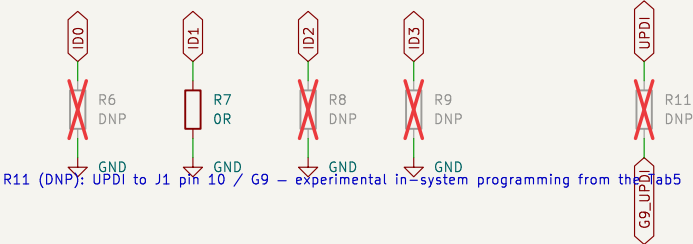
UPDI programming (SerialUPDI / MPLAB Snap)



Decoupling (VDD x2, VDDIO2), I2C pull–ups, RESET pull–up

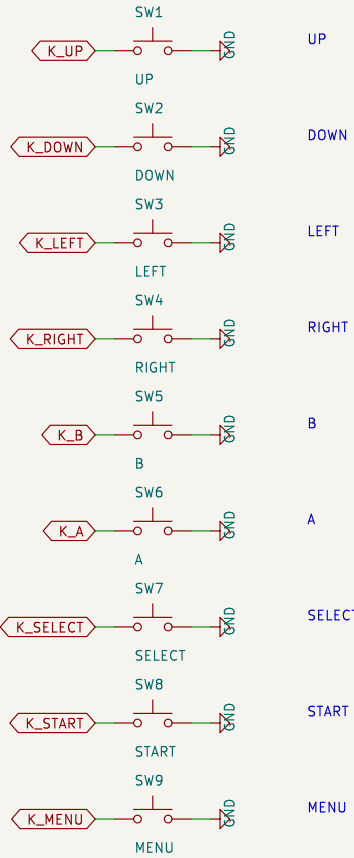
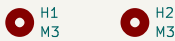


Console ID straps: 0R to GND = bit set. Game Boy = ID 2 (fit R7; R6, R8, R9 DNP)



R11 (DNP): UPDI to J1 pin 10 / G9 – experimental in–system programming from the Tab5

Mounting holes (M3, 96 mm apart)



Tab5 keyboard–port board: AVR32DD28 + firmware_avrdd, I2C 0x6D
Generated by kicad_gen/gen_sch.py from gen_pcb.py and layout.py

RetroESP32–P4

Sheet: /
File: retropad_gb.kicad_sch

Title: RetroPad Game Boy controller (console ID 2)

Size: A3 Date: 2026–10–05
KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4

Rev: 0.1
Id: 1/1